

- 4 (a) State what is meant by the frequency of the oscillations of an oscillating object.

.....
 [1]

- (b) An object is oscillating.

Fig. 4.1 shows the variation of the acceleration a of the object with its displacement x from the equilibrium position.

Fig. 4.2 shows the variation of the kinetic energy E_K of the object with time t .

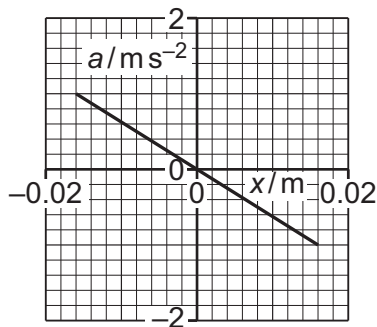


Fig. 4.1

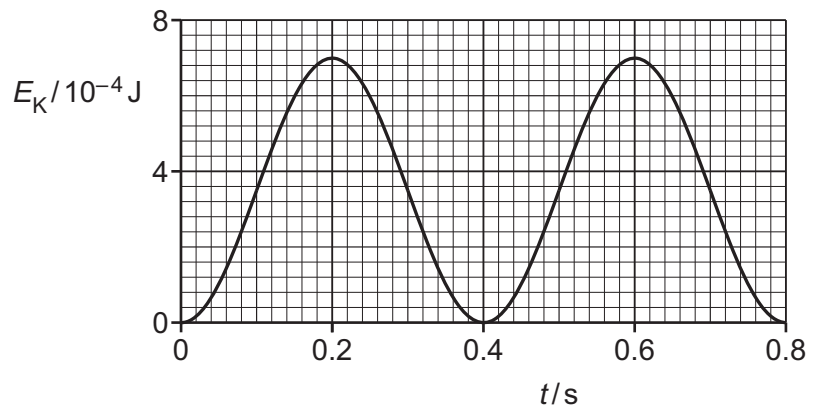


Fig. 4.2

- (i) Explain how Fig. 4.2 shows that the period of the oscillations is 0.80 s.

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 [1]

- (ii) Calculate the angular frequency ω of the oscillations.

$\omega = \dots\dots\dots \text{rad s}^{-1}$ [2]

